Practical 4a: PWA with Web App Manifest and Push Notifications

AIM: To create a PWA with a complete Web App Manifest file (app name, icons, start URL, display mode)and implement Push Notifications.

manifest.json

```
{
"name": "MyApp - Full Featured PWA",
"short_name": "MyApp",
"description": "A fully featured Progressive Web App",
"start_url": "/index.html?source=pwa",
"scope": "/",
"display": "standalone",
"orientation": "portrait-primary",
"background_color": "#1F4E79",
"theme_color": "#2E75B6",
"lang": "en-US",
"categories": ["education", "utilities"],
"icons": [
{ "src": "/icons/icon-72x72.png", "sizes": "72x72", "type": "image/png" },
{ "src": "/icons/icon-96x96.png", "sizes": "96x96", "type": "image/png" },
{ "src": "/icons/icon-128x128.png", "sizes": "128x128", "type": "image/png" },
{ "src": "/icons/icon-192x192.png", "sizes": "192x192", "type": "image/png",
"purpose": "any maskable" },
{ "src": "/icons/icon-512x512.png", "sizes": "512x512", "type": "image/png",
"purpose": "any maskable" }
],
"shortcuts": [
{
"name": "View Students",
"url": "/students",
"icons": [{ "src": "/icons/students.png", "sizes": "96x96" }]
}
]
}
```

push-notification.js

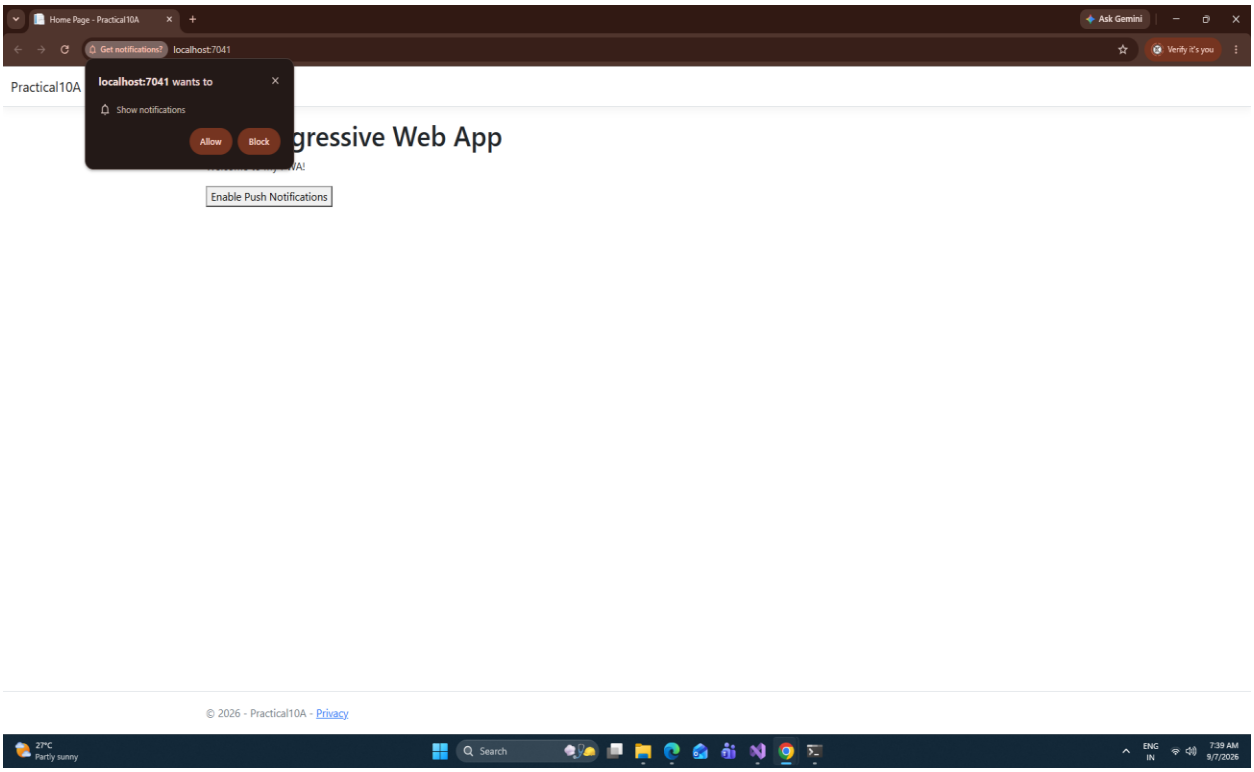
push-notification.js

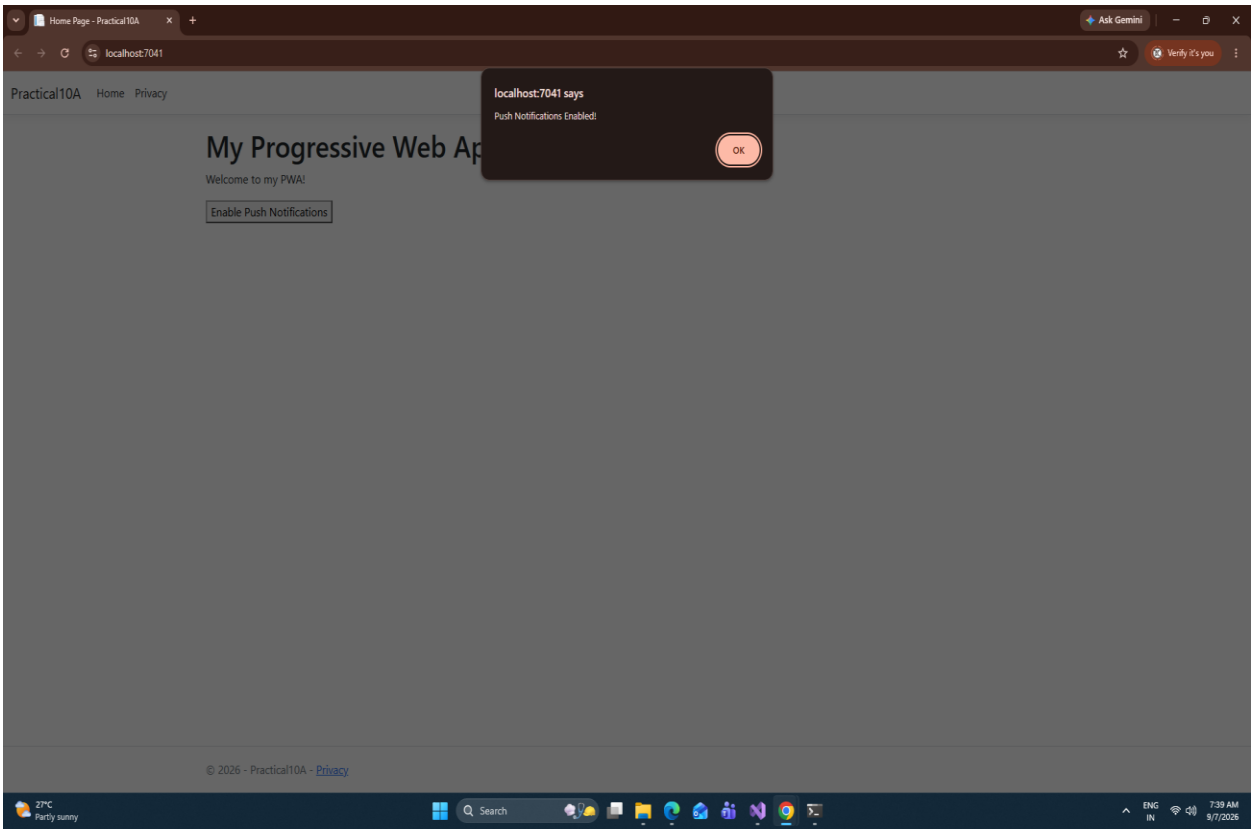
```
// Request notification permission
async function requestPermission() {
```

```
const permission = await Notification.requestPermission();
if (permission === 'granted') {
console.log('Notification permission granted');
subscribeUser();
} else {
console.log('Notification permission denied');
}
}
// Subscribe to push notifications
async function subscribeUser() {
const registration = await navigator.serviceWorker.ready;
const subscription = await registration.pushManager.subscribe({
userVisibleOnly: true,
applicationServerKey: urlBase64ToUint8Array('YOUR_VAPID_PUBLIC_KEY')
});
console.log('Push subscription:', JSON.stringify(subscription));
// Send subscription to server
await fetch('/api/subscribe', {
method: 'POST',
headers: { 'Content-Type': 'application/json' },
body: JSON.stringify(subscription)
});
}
// Show local notification demo
function showNotification() {
if (Notification.permission === 'granted') {
new Notification('MyApp Notification', {
body: 'You have a new message!',
icon: '/icons/icon-192x192.png',
badge: '/icons/badge.png',
tag: 'demo-notification'
});
}
}
// Service Worker: push event handler
// (in service-worker.js)
self.addEventListener('push', event => {
const data = event.data ? event.data.json() : {};
const options = {
body: data.body || 'New notification!',
```

```
icon: '/icons/icon-192x192.png',
badge: '/icons/badge.png',
vibrate: [200, 100, 200]
};
event.waitUntil(
self.registration.showNotification(data.title || 'MyApp', options)
);
});
```

Output




Home Page - Practical10A
localhost:7041
Ask Gemini
Practical10A
Home
Privacy
localhost:7041 says
Push Notifications Enabled!
OK
My Progressive Web Ap
Welcome to my PWA!
Enable Push Notifications
© 2026 - Practical10A - Privacy
27°C
Partly sunny
Search
ENG
IN
7:39 AM
9/7/2026